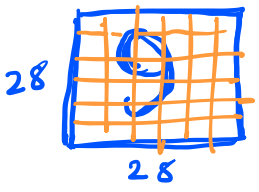
Neural-networks - part-3.

Design the Network. Understand every update.
Train with intent.

MNIST problem:

70k, handwritten digit images → 0-9 range



multiclass → 10 classes.

flatten → 1 x 784 → 28x28

$$X_{scaled} = \frac{X}{255} \rightarrow range \in [0, 1]$$

Architecture:

(784, 256, 128, 64, 10).

(784, 1024, 1024, 1024, 10)

(784, 128, 64, 10)

(784, 128, 10)

Inference

model.predict_proba (val'd)

→ probs for each class.

we take argmax.

Optimizers

① Stochastic Gradient Descent:

$$w_t = w_{t-1} - \eta \cdot g_t$$

- w_t: new weight
- w_{t-1}: old weight
- η: learning rate
- g_t: gradient

$$L(w) = (w-3)^2, \quad \frac{\partial L}{\partial w} = 2(w-3)$$

$w_0 = 0, \quad \eta = 0.1$

$w_0 = 0, \quad g_1 = 2(0-3) = -6, \quad w_1 = 0 - 0.1 \times (-6) = 0.6$

$w_1 = 0.6, \quad g_2 = 2(0.6-3) = -4.8,$

$w_2 = 0.6 - 0.1 \times (-4.8) = 1.08$

$L(w_0) = 9, \quad L(w_1) = 5.76, \quad L(w_2) = 3.68$

$$L(w_1, w_2) = 0.1 w_1^2 + 2 w_2^2$$

Least loss will be when $w_1 = w_2 = 0$.

$L(w_1, w_2) = 0$

$$\nabla L(w_1, w_2) = \begin{bmatrix} 0.2 w_1 \\ 4 w_2 \end{bmatrix}$$

SGD with momentum:

$$v_t = \beta v_{t-1} + g_t, \quad v_t = 0$$

- v_t: velocity
- β: momentum co-efficient

$$w_t = w_{t-1} - \eta \cdot v_t$$

* Adam (Adaptive momentum)

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \qquad m_t = 0$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\, g_t \odot g_t$$

$\odot$ → Element wise mult.

$$m_t = \frac{m_t}{1-\beta_1^t} \,, \quad v_t = \frac{v_t}{1-\beta_2^t}$$

$$w_t = w_{t-1} - \eta \frac{m_t}{\sqrt{v_t} + \epsilon}$$